

Week 1: Literature Review

Analysing the Role of Big Data Analytics for Cyber Threat Detection and Prevention in Banking Organisations of Ireland

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Course: Open Data Practice - Research Practicum

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Document: Week 1 - Literature Review (2025-2026)

Papers: 15 Peer-Reviewed Papers

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Each paper is reviewed under: What is the Paper | What Does It Do | Research Gap | Future Findings | Baseline Status

1. Introduction

This Week 1 literature review covers 15 peer-reviewed papers published between 2025 and 2026, directly relevant to the research on Big Data Analytics for Cyber Threat Detection and Prevention in Irish Banking Organisations. The review is structured across five thematic areas:

- (a) Big Data Analytics in Cybersecurity (Papers 1-3)
- (b) Machine Learning for Cyber Threat Detection (Papers 4-6)
- (c) Intrusion Detection Systems - IDS (Papers 7-9)
- (d) Fraud Detection in Banking Using ML (Papers 10-12)
- (e) Predictive Analytics and Cyber Threat Intelligence (Papers 13-15)

For each paper, four key questions are answered:

- 1. What is the Paper? - The scope and context of the study
- 2. What Does It Do? - The methodology, techniques and results
- 3. Research Gap - What the paper identified as missing in existing literature
- 4. Future Findings - Recommendations and future research directions

Papers marked with [BASELINE] are designated as BASELINE PAPERS whose methodology is directly adopted or extended in the current study.

Section A: Big Data Analytics in Cybersecurity (Papers 1-3)

Paper 1 [BASELINE] BASELINE PAPER

BankNet: Real-Time Big Data Analytics for Secure Internet Banking

Authors: Sathupadi, K., Achar, S., Bhaskaran, S.V. et al. | Year: 2025 | Journal: *Big Data and Cognitive Computing (MDPI)*, Vol. 9, No. 2, p. 24

URL: <https://www.mdpi.com/2504-2289/9/2/24>

What is the Paper?:

This paper introduces BankNet, a real-time big data analytics framework designed specifically for securing internet banking services. It addresses the growing need for intelligent, scalable security infrastructures in digital banking environments.

What Does It Do?:

BankNet integrates Apache Kafka-based streaming data pipelines with machine learning classifiers to detect anomalous banking transactions in real time. The framework processes large volumes of transaction data and triggers automated alerts when suspicious patterns are detected. It was tested on simulated banking transaction datasets and achieved sub-second threat detection latency.

Research Gap Addressed:

Existing big data security frameworks were not purpose-built for the unique characteristics of banking data such as high transaction velocity, varied transaction types, and strict regulatory environments. Most general-purpose frameworks lacked real-time capabilities for banking-specific use cases.

Future Findings / Recommendations:

Future work should incorporate federated learning to allow multi-bank collaboration without sharing sensitive transaction data. The authors also recommend integration with explainable AI (XAI) tools to satisfy regulatory transparency requirements in Europe and Ireland.

[BASELINE] BASELINE PAPER: This paper serves as a key baseline for the current study. Its methodology, dataset approach, or model comparison framework is directly adopted and extended.

Paper 2

Big Data in Cybersecurity: Enhancing Threat Detection with AI and ML

Authors: Kumar, B.H., Nuka, S.T., Malempati, M. et al. | Year: 2025 | Journal: *Metallurgical and Materials Engineering*

URL: <https://www.academia.edu/download/122135269/2.pdf>

What is the Paper?:

This paper is a broad review study that examines the intersection of big data technologies and cybersecurity, assessing how AI and ML can enhance threat detection capabilities in large-scale digital environments.

What Does It Do?:

The paper evaluates big data frameworks including Hadoop, Spark, and Flink for their suitability in cybersecurity applications. It also reviews ML algorithms such as Random Forest, SVM, and neural networks, comparing their accuracy and scalability when applied to large security datasets. The study concludes that big data-driven ML approaches outperform traditional rule-based systems by up to 40%.

Research Gap Addressed:

The paper identifies that most existing cybersecurity solutions were built for small-to-medium scale data and struggle with the volume, velocity, and variety of modern cyber threat data. There is limited research on integrating big data frameworks with lightweight, real-time ML models suitable for production banking environments.

Future Findings / Recommendations:

Research should focus on developing lightweight ML models that can operate efficiently within big data pipelines without sacrificing detection accuracy. The authors also suggest applying transfer learning to reduce retraining costs when new threat patterns emerge.

Paper 3

Enhancing Cybersecurity Risk Management in Fintech Through Advanced Analytics and Machine Learning

Authors: Kokogho, E., Okon, R., Omowole, B.M. et al. | Year: 2025 | Journal: Frontiers in Science

URL: <https://www.researchgate.net/publication/388459015>

What is the Paper?:

This paper focuses on cybersecurity risk management in fintech organisations, which face unique security threats due to their heavy reliance on digital platforms, APIs, and third-party integrations. It proposes an advanced analytics framework for proactive risk management.

What Does It Do?:

The study builds a multi-layered cybersecurity risk management framework that uses predictive analytics and ML classifiers to score and prioritise cyber risks. It applies the framework to real fintech transaction data and demonstrates a 35% improvement in early threat identification compared to conventional risk scoring systems.

Research Gap Addressed:

Most cybersecurity risk management research has focused on large traditional banks, leaving a gap for fintech-specific frameworks. Additionally, existing risk scoring models are largely static and fail to adapt to evolving threat landscapes dynamically.

Future Findings / Recommendations:

The authors recommend the development of self-adapting risk management models that update continuously based on new threat intelligence. Integration with open banking APIs and regulatory frameworks like PSD2 in the EU/Ireland is also identified as a key future direction.

Section B: Machine Learning for Cyber Threat Detection (Papers 4-6)

Paper 4 [BASELINE] BASELINE PAPER

Machine Learning-Driven Approaches to Cyber Threat Detection: Enhancing IoT Security

Authors: Ali, M., Raza, A., Akram, M.A., Arif, H., Ali, A. | Year: 2025 | Journal: Journal of Informatics and IT (acsit.org)

URL: <https://www.acsit.org/index.php/jiite/article/download/64/41>

What is the Paper?:

This paper investigates how machine learning-driven approaches can be used to detect cyber threats in IoT-connected environments. It is relevant to banking because modern banks deploy IoT devices for ATMs, point-of-sale terminals, and smart branch infrastructure.

What Does It Do?:

The study evaluates Random Forest, SVM, Decision Tree, and gradient boosting algorithms for classifying network traffic in IoT environments into normal and attack categories. Tested on the CICIDS 2017 dataset, Random Forest achieved the highest accuracy of 98.4% with an F1-Score of 0.97. The paper also demonstrates that feature selection reduces model training time by 28%.

Research Gap Addressed:

Prior ML-based threat detection research focused on traditional network environments and did not adequately address the resource-constrained nature of IoT devices used in banking infrastructure. There is a lack of lightweight ML models that perform well on both accuracy and computational efficiency.

Future Findings / Recommendations:

Future work should develop edge-deployable ML models that can run directly on IoT devices within banking environments. The authors also recommend using federated learning to train models across multiple banking branches without centralising sensitive data.

[BASELINE] BASELINE PAPER: This paper serves as a key baseline for the current study. Its methodology, dataset approach, or model comparison framework is directly adopted and extended.

Paper 5

Machine Learning-Based Anomaly Detection for Cyber Threat Prevention

Authors: Mizanur, M., Kumer, S., Reza, N. | Year: 2025 | Journal: Journal of Primeasia (emanresearch.org)

URL: https://publishing.emanresearch.org/CurrentIssuePDF/EmanPublisher_26_5955primeasia-6110172.pdf

What is the Paper?:

This study compares supervised and unsupervised machine learning approaches for anomaly-based cyber threat detection. It is a well-cited paper (37 citations) that provides a solid comparison framework applicable to this research.

What Does It Do?:

The paper tests six ML algorithms - Isolation Forest, Autoencoder, K-Means (unsupervised) and Random Forest, SVM, Naive Bayes (supervised) - on a combined network traffic dataset. The results show that hybrid approaches combining unsupervised anomaly scoring with supervised classification achieve the best precision (0.96) and recall (0.94) values. The study also analyses the impact of class imbalance on model performance.

Research Gap Addressed:

Most existing anomaly detection research uses either fully supervised or fully unsupervised approaches. There is

limited research on hybrid frameworks that leverage the strengths of both paradigms for cyber threat detection, particularly in financial data contexts.

Future Findings / Recommendations:

The authors recommend applying hybrid anomaly detection frameworks to banking transaction data where labelled fraud cases are scarce. Future research should also explore online learning models that continuously adapt to new types of anomalies in real time.

Paper 6

Adversarial Machine Learning Framework for Robust Banking Security: The ACTP Tool

Authors: Padmanaban, H., Sharma, Y.K., Sharma, P. et al. | Year: 2026 | Journal: *Journal of The Institution of Engineers (India)*, Springer

URL: <https://link.springer.com/article/10.1007/s40031-026-01337-1>

What is the Paper?:

This 2026 paper presents the Automated Cyber Threat Detection and Prevention (ACTP) tool, a banking-specific adversarial ML framework that is one of the most directly relevant papers to this study due to its exclusive focus on banking cybersecurity.

What Does It Do?:

The ACTP tool uses adversarial training to make ML-based cyber threat detectors robust against evasion attacks. The framework tests Random Forest and SVM classifiers under adversarial conditions and shows that adversarial training improves model robustness by 42% without significantly reducing standard detection accuracy. The tool is evaluated on synthetic banking transaction attack scenarios.

Research Gap Addressed:

Existing banking security ML models were shown to be highly vulnerable to adversarial examples where attackers craft fraudulent transactions that bypass ML detectors. This robustness gap had not been addressed in prior banking ML security literature.

Future Findings / Recommendations:

The authors call for real-world deployment trials of adversarially trained models in production banking environments. They also recommend extending the framework to include deep learning models and testing against a wider range of adversarial attack strategies.

Section C: Intrusion Detection Systems - IDS (Papers 7-9)

Paper 7 [BASELINE] BASELINE PAPER

A Comprehensive Survey on Intrusion Detection Systems with Advances in ML, DL and Emerging Cybersecurity Challenges

Authors: Hozouri, A., Mirzaei, A., Effatparvar, M. | Year: 2025 | Journal: Discover Artificial Intelligence, Springer

URL: <https://link.springer.com/article/10.1007/s44163-025-00578-1>

What is the Paper?:

This is the most highly cited paper in this review (181 citations) and provides a comprehensive survey of intrusion detection systems (IDS) from traditional rule-based approaches to modern ML and deep learning methods. It is an authoritative reference for IDS research.

What Does It Do?:

The survey analyses over 150 IDS papers and categorises them by detection method (signature-based, anomaly-based, hybrid), ML algorithm used, and dataset benchmarked. It concludes that Random Forest and CNN-LSTM hybrid models achieve the highest detection rates across standard benchmarks including NSL-KDD, UNSW-NB15, and CICIDS2017. The paper also discusses concept drift and adversarial robustness as key unresolved challenges.

Research Gap Addressed:

The survey identifies a significant gap in real-world IDS deployment studies. Most research evaluates models on benchmark datasets that do not reflect current threat landscapes. There is also a lack of IDS research specifically targeting financial and banking network environments.

Future Findings / Recommendations:

The authors recommend developing domain-specific IDS solutions for high-value targets like banking networks. Future research should focus on adaptive IDS that can detect previously unseen attack patterns using zero-shot or few-shot learning techniques.

[BASELINE] BASELINE PAPER: This paper serves as a key baseline for the current study. Its methodology, dataset approach, or model comparison framework is directly adopted and extended.

Paper 8

An Advanced Cyber Security Model Using Federated Machine Learning Approach for Intrusion Detection in Networks

Authors: Laddi, M., Allagi, S., Rachh, R. et al. | Year: 2025 | Journal: Journal of Cloud Computing and Engineering (bonviewpress.com)

URL: <https://ojs.bonviewpress.com/index.php/JCCE/article/download/3751/1215>

What is the Paper?:

This paper proposes a federated machine learning model for intrusion detection that allows multiple organisations to collaboratively train an IDS without sharing raw sensitive data. This is particularly relevant in GDPR-regulated environments like Ireland.

What Does It Do?:

The federated IDS trains local models at each participating network node and aggregates model updates centrally

without transmitting raw traffic data. The study shows that the federated model achieves within 2-3% of the accuracy of a centrally trained model while preserving complete data privacy. DDoS and data exfiltration attacks were detected with 95.6% accuracy.

Research Gap Addressed:

Traditional IDS models require centralised data collection, which conflicts with GDPR and banking data protection regulations. There was no established federated IDS framework tested in multi-organisational banking environments before this study.

Future Findings / Recommendations:

The authors highlight the need for federated IDS standards that could be adopted by banking consortia in Ireland and the EU. Future work should also address security vulnerabilities in the federated aggregation process itself, such as model poisoning attacks.

Paper 9

Development of an Intelligent Intrusion Detection System for IoT Networks Using Deep Learning

Authors: Zhang, H. | Year: 2025 | Journal: Discover Internet of Things, Springer

URL: <https://link.springer.com/article/10.1007/s43926-025-00177-7>

What is the Paper?:

This paper develops a deep learning-based IDS for IoT networks, addressing the security of interconnected smart devices - including banking ATMs, point-of-sale terminals, and connected branch systems - which are increasingly targeted by cyber attackers.

What Does It Do?:

The study uses a CNN-LSTM architecture to analyse sequential IoT network traffic data for intrusion detection. The model is trained on the CIC-IoT2023 dataset and achieves 99.1% accuracy with a false positive rate below 0.4%. The paper demonstrates that temporal feature extraction through LSTM significantly improves detection of slow-moving attacks like APTs.

Research Gap Addressed:

Most deep learning IDS research was conducted on traditional network datasets, not IoT-specific data. There was limited work on detecting advanced persistent threats (APTs) in IoT-connected banking infrastructure using temporal deep learning models.

Future Findings / Recommendations:

Future research should focus on deploying lightweight versions of CNN-LSTM models on IoT edge devices with limited computing power. The author also recommends creating a banking-specific IoT threat dataset, as current public datasets do not represent financial IoT environments.

Section D: Fraud Detection in Banking Using Machine Learning (Papers 10-12)

Paper 10 [BASELINE] BASELINE PAPER

Machine Learning-Based Fraud Detection Systems and Their Effectiveness in Reducing Cybersecurity Risks in Digital Banking

Authors: Naeem, W., Butt, M.A., Javeid, U. | Year: 2026 | Journal: Social Science Review Archives

URL: <https://policyjournalofms.com/index.php/6/article/download/2344/2266>

What is the Paper?:

This 2026 paper directly evaluates machine learning-based fraud detection systems in digital banking, comparing Decision Tree, Random Forest, and SVM - the exact three models used in the current study. It is therefore the most directly relevant paper in this entire review.

What Does It Do?:

The paper applies all three models to a digital banking transaction dataset and reports accuracy, precision, recall, F1-score and ROC-AUC for each. Random Forest achieves the best overall performance (Accuracy: 99.1%, AUC: 0.987), followed by SVM (Accuracy: 97.8%, AUC: 0.961) and Decision Tree (Accuracy: 96.2%, AUC: 0.934). SMOTE is used to handle class imbalance in the training data.

Research Gap Addressed:

Previous fraud detection studies focused on credit card fraud using general financial datasets rather than comprehensive digital banking transaction data. There was a lack of comparative studies evaluating these three specific models together on banking cybersecurity datasets with proper class imbalance handling.

Future Findings / Recommendations:

Future work should explore explainable AI techniques to make Random Forest predictions interpretable for bank compliance officers. The authors also recommend extending the evaluation to real-time streaming environments where models must make predictions on transaction data within milliseconds.

[BASELINE] BASELINE PAPER: This paper serves as a key baseline for the current study. Its methodology, dataset approach, or model comparison framework is directly adopted and extended.

Paper 11

Early Detection of Digital Transaction Fraud in Banking Systems Using the Random Forest Algorithm

Authors: Dhammayogi, M.B.D., Pramatha, C.R.A. | Year: 2026 | Journal: Journal of Information Technology and Computer Science (JITECS)

URL: <https://jitecs.ub.ac.id/index.php/jitecs/article/view/912>

What is the Paper?:

This paper focuses specifically on the early detection of digital transaction fraud using the Random Forest algorithm, providing a detailed analysis of how Random Forest can be optimised for banking fraud detection scenarios.

What Does It Do?:

The study conducts an in-depth evaluation of Random Forest with varying numbers of decision trees (50 to 500) on a banking transaction dataset. It finds that 200 trees provides the optimal balance between accuracy (98.7%) and training time. Feature importance analysis identifies transaction amount, time of day, and transaction frequency as the top three predictive features for fraud.

Research Gap Addressed:

While Random Forest is widely used in fraud detection, there was limited analysis of hyperparameter optimisation for banking-specific datasets. Most prior studies used default settings without systematically evaluating the impact of tree count and feature selection on detection performance.

Future Findings / Recommendations:

The authors recommend future studies to compare Random Forest with emerging gradient boosting methods like XGBoost and LightGBM in the same banking context. They also suggest testing early detection capabilities using streaming data platforms like Apache Kafka for real-time fraud alerts.

Paper 12

Enhanced Banking Fraud Detection: A Comparative Analysis of Supervised Machine Learning Algorithms

Authors: Sanda, A.M., Mukhtar, U.S. et al. | Year: 2026 | Journal: Proceedings of Engineering and Technology Innovation (FETICON)

URL: <https://proceedings.feticon.com.ng/index.php/feticon/article/view/332>

What is the Paper?:

This 2026 paper presents an enhanced comparative study of multiple supervised ML algorithms for banking fraud detection, including the three algorithms central to this study: Decision Tree, Random Forest, and SVM.

What Does It Do?:

The study compares six algorithms - Decision Tree, Random Forest, SVM, Logistic Regression, KNN, and Naive Bayes - on a banking fraud dataset. Evaluation uses Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Random Forest consistently outperforms all others, while Decision Tree offers the most interpretable results. SVM performs well on high-dimensional feature spaces but has higher computational cost.

Research Gap Addressed:

Prior comparative studies of ML algorithms for banking fraud detection lacked consistency in evaluation metrics and dataset selection. There was no recent comprehensive comparison specifically targeting digital banking fraud with modern, large-scale datasets post-2023.

Future Findings / Recommendations:

Future research should evaluate deep learning models such as LSTM and Transformer architectures as next-generation alternatives to the three classical models. The authors also recommend developing hybrid models that combine the interpretability of Decision Tree with the accuracy of Random Forest through cost-sensitive learning.

Section E: Predictive Analytics and Cyber Threat Intelligence (Papers 13-15)

Paper 13

The Role of Threat Intelligence in Preventing Financially Motivated Cyberattacks

Authors: Eshra, S.A., Zohora, F.T., Akter, S., Rasul, I. et al. | Year: 2025 | Journal: *Cyber Intelligence Review (jecir.com)*

URL: <https://jecir.com/index.php/jecir/article/view/24>

What is the Paper?:

This paper examines how cyber threat intelligence (CTI) can be used proactively to prevent financially motivated cyberattacks targeting banking and financial institutions. It bridges the gap between threat intelligence research and practical banking security applications.

What Does It Do?:

The study develops a CTI-driven security framework that ingests threat intelligence feeds from sources like MITRE ATT&CK and OSINT platforms, and uses ML classifiers to prioritise and respond to threat indicators. The framework reduced mean time to detect (MTTD) by 58% and mean time to respond (MTTR) by 43% in a banking security operations centre (SOC) simulation.

Research Gap Addressed:

Most CTI research focused on general enterprise security rather than the specific threat landscape facing financial institutions. There was a gap in research on operationalising threat intelligence within banking SOCs using automated ML-driven analysis.

Future Findings / Recommendations:

The authors recommend integrating CTI frameworks with real-time big data analytics pipelines in banking systems. They also call for industry-wide threat intelligence sharing platforms for Irish and EU financial institutions to improve collective cyber defence.

Paper 14

Risk Assessment Framework for Banking Institutions: Integrating Predictive Financial Analytics and Cybersecurity Risk Intelligence

Authors: Ullah, N., Ishaq, M., Rahman, F., Aslam, M.U. et al. | Year: 2026 | Journal: *Journal of Management Science Research Reports*

URL: <https://www.jmsrr.com/index.php/Journal/article/view/708>

What is the Paper?:

This 2026 paper proposes a comprehensive risk assessment framework for banking institutions that combines predictive financial analytics with cybersecurity risk intelligence. It takes a holistic approach linking financial risk modelling with cyber threat modelling.

What Does It Do?:

The framework uses predictive analytics models trained on historical financial and cybersecurity event data to forecast and quantify cyber risk in monetary terms. It integrates regulatory compliance requirements and demonstrates that predictive cyber risk scores can reduce unexpected cybersecurity-related financial losses by up to 31% compared to traditional audit-based risk assessment approaches.

Research Gap Addressed:

The financial and cybersecurity risk assessment domains have historically been treated separately by banking institutions. There was a significant gap in integrated frameworks that can quantify cybersecurity risks in financial terms aligned with banking regulatory requirements.

Future Findings / Recommendations:

The authors recommend expanding the framework to incorporate Irish and EU-specific regulatory requirements including DORA (Digital Operational Resilience Act) and NIS2 Directive. Future studies should also test the framework's predictive accuracy using live banking data from central banking institutions.

Paper 15

Artificial Intelligence and Cybersecurity in the Banking Sector: Opportunities and Risks

Authors: Kovacevic, A., Radenkovic, S. et al. | Year: 2025 | Journal: *Contemporary Issues in Economics and Business*, University of Belgrade

URL: https://rhinosec.fb.bg.ac.rs/bitstream/handle/123456789/491/bitstream_1162.pdf

What is the Paper?:

This paper provides a critical analysis of the opportunities and risks associated with deploying artificial intelligence and big data analytics for cybersecurity in the banking sector. It offers a balanced strategic perspective that is valuable for formulating recommendations in this study.

What Does It Do?:

The study reviews AI-powered cybersecurity tools deployed in banking organisations and assesses their effectiveness against common financial cyber threats. It identifies that AI-driven security tools reduce fraud losses by an average of 52% but also introduces new risks including adversarial attacks, AI model bias, and over-reliance on automated systems. The paper provides strategic recommendations for responsible AI deployment in banking.

Research Gap Addressed:

Most AI cybersecurity research presents only the benefits of AI adoption without critically examining the risks. There is a gap in balanced research that helps banking decision-makers understand both the potential and limitations of AI-driven security in regulated banking environments.

Future Findings / Recommendations:

The authors recommend developing AI governance frameworks specifically for banking cybersecurity, including model transparency standards, bias auditing, and adversarial testing protocols. They emphasise that Irish banking organisations should align AI cybersecurity strategies with the EU AI Act requirements currently being enforced.

Summary: Baseline Papers and Their Role in This Study

Three papers have been designated as BASELINE PAPERS for the current study. These papers directly inform the methodology, model selection, evaluation strategy, and research design.

[BASELINE] Paper 1 - Sathupadi et al. (2025) - BankNet

Journal: B

Role in This Study: BankNet's real-time big data analytics framework provides the architectural blueprint for the data processing pipeline used in this study. Its approach to integrating streaming transaction data with ML classifiers is adopted as the foundational system design. This study extends BankNet by applying it to a specific Irish banking cybersecurity dataset and comparing three ML models.

[BASELINE] Paper 4 - Ali et al. (2025) - ML-Driven Cyber Threat Detection

Journal: J

Role in This Study: This paper provides the comparative ML evaluation framework used as the methodological baseline. Its approach to benchmarking Random Forest, SVM, and Decision Tree on security datasets using Accuracy, Precision, Recall, F1-Score, and ROC-AUC is directly replicated in this study. The feature selection methodology from this paper is also adapted for the banking transaction dataset.

[BASELINE] Paper 7 - Hozouri et al. (2025) - Comprehensive IDS Survey

Journal: D

Role in This Study: As the most cited paper in this review (181 citations), this survey establishes the theoretical grounding for this study's use of anomaly-based IDS principles. The three ML models selected for this study (Decision Tree, Random Forest, SVM) are validated by this survey as the most appropriate classical ML approaches for cyber threat detection in structured financial data.

[BASELINE] Paper 10 - Naeem et al. (2026) - ML Fraud Detection in Digital Banking

Journal: S

Role in This Study: This is the most directly comparable study to the current research. It applies the same three ML models (Decision Tree, Random Forest, SVM) to a digital banking fraud detection task and uses identical evaluation metrics. The performance benchmarks reported in this paper serve as the comparison baseline against which the current study's results will be measured and discussed.

Master Summary Table: All 15 Papers

#	Authors	Year	Journal	Key Contribution	Gap / Future Work
1	Sathupadi et al.	2025	Big Data Cog. Comp.	Real-time banking security (BankNet) [BL]	Federated learning; XAI integration
2	Kumar et al.	2025	Metallurgical Mat. Eng.	Big data + AI/ML threat detection review	Lightweight real-time models needed
3	Kokogho et al.	2025	Frontiers in Science	Fintech cybersecurity risk mgmt framework	Self-adapting models; PSD2 integration
4	Ali et al.	2025	J. Informatics IT	ML comparison IoT threat detection [BL]	Edge-deployable banking IoT models
5	Mizanur et al.	2025	J. Primeasia	Hybrid anomaly detection framework	Banking with scarce fraud labels
6	Padmanaban et al.	2026	Springer J. Inst. Eng.	Adversarial ML banking (ACTP tool)	Real-world deploy; DL extension
7	Hozouri et al.	2025	Discover AI Springer	IDS survey ML/DL (181 cites) [BL]	Domain-specific banking IDS
8	Laddi et al.	2025	J. Cloud Comp. Eng.	Federated ML GDPR-compliant IDS	Federated banking IDS standards
9	Zhang, H.	2025	Discover IoT Springer	CNN-LSTM DL IDS for IoT networks	Banking IoT threat dataset needed
10	Naeem et al.	2026	Social Sci. Rev. Arch.	DT vs RF vs SVM banking fraud [BL]	XAI; real-time streaming eval.
11	Dhammayogi et al.	2026	JITECS	RF hyperparameter optimisation fraud	Compare XGBoost; streaming eval
12	Sanda et al.	2026	FETICON Proc.	Comparative supervised ML banking fraud	DL and hybrid cost-sensitive models
13	Eshra et al.	2025	Cyber Intel. Review	CTI framework financial cyberattacks	EU/Ireland threat intel. sharing
14	Ullah et al.	2026	J. Mgmt Sci. Res.	Predictive analytics + cyber risk intel.	DORA/NIS2; central bank data test
15	Kovacevic et al.	2025	Univ. of Belgrade	AI opportunities/risks banking security	EU AI Act; AI governance frameworks

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